

HDOC Day 31

Problem

Write a C program to perform the following menu-driven operations on a given 2D array:

1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit

C code

```
#include <stdio.h>

int main()
{
    int ch;

    printf("choice 1 is to find the sum of elements of 2d array\n");
    printf("choice 2 is to find the sum of each row of 2d array and store it in an array\n");
    printf("choice 3 is to find the transpose of a 2d array matrix\n");
    printf("exit is choice 4\n");
    printf("enter your choice\n");
    scanf("%d",&ch);
    switch (ch)
    {
        case 1:
        {
            int n,m;

            printf("enter the size of array(row*column)\n");
            scanf("%d%d",&n,&m);
```

```

int arr[n][m];

printf("enter the elements of the array\n");

for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
        scanf("%d",&arr[i][j]);
}

int sum=0;

for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        sum=sum+arr[i][j];
    }
}

printf("the sum of array elements is%d\n",sum);

break;
}

```

case 2:

```

{
    int n,m;

    printf("enter the size of array(row*column)\n");

    scanf("%d%d",&n,&m);

    int arr1[n][m];

    for(int i=0;i<n;i++)
    {
        for(int j=0;j<m;j++)
            scanf("%d",&arr1[i][j]);
    }
}

```

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    }

    int arr2[n];

    printf("matrix is:\n");

    for(int i=0;i<n;i++)

    {int sum=0;

        for(int j=0;j<m;j++)

        {

            printf("%d\t",arr1[i][j]);

            sum=sum+arr1[i][j];

            arr2[i]=sum;

        }

        printf("\n");

    }

    printf("the new array made of the sum of the elements of each row of matrix is\n");

    for(int i=0;i<n;i++)

    {

        printf("%d\t",arr2[i]);

    }

    break;

}

case 3:

{

    int n,m;

    printf("enter the size of array(row*column)\n");

    scanf("%d%d",&n,&m);

    int arr1[n][m];

    int arr2[m][n];

    printf("array elements input\n");

```

```

for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        scanf("%d",&arr1[i][j]);
    }
}
printf("original array\n");
for(int i=0;i<n;i++)
{
    for(int j=0;j<m;j++)
    {
        printf("%d\t",arr1[i][j]);
    }
    printf("\n");
}
for(int i=0;i<m;i++)
{
    for(int j=0;j<n;j++)
    {
        arr2[i][j]=arr1[j][i];
    }
}

printf("\nthe new transpose array is\n");
for(int i=0;i<m;i++)
{
    for(int j=0;j<n;j++)
    {

```

```
        printf("%d\t",arr2[i][j]);
    }
    printf("\n");
}
break;
}
case 4:
{
    printf("exit");
    return 0;
}
default:
{
    printf("invalid choice\n");
    break;
}
}
return 0;
}
```